# Zikun Wang

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## Education

**Northeastern University** — Boston, MA
*PhD in Computer Science* — *Sep. 2025*

**Boston University** — Boston, MA
*Bachelor of Arts and Master of Science in Computer Science* — *Jan. 2021 – Dec 2024*

## Conference Publications

[1] **The Dangerous Impact of Solver Imprecision on Data Management Techniques (and How to Avoid It).** Zixuan Chen, Zikun Wang, Panagiotis Manolios, and Mirek Riedewald. In *Proceedings of the 30th International Conference on Extending Database Technology (EDBT 2027).*

[2] **GPU-Accelerated ANNS: Quantized for Speed, Built for Change.** Hunter McCoy, Zikun Wang, and Prashant Pandey. In *Proceedings of the VLDB Endowment (VLDB 2026).*

[3] **CAPSys: Contention-Aware Task Placement for Data Stream Processing.** Yuanli Wang, Lei Huang, Zikun Wang, Vasiliki Kalavri, and Ibrahim Matta. In *Proceedings of the Twentieth European Conference on Computer Systems (EuroSys 2025).*

## Experience

**Research Assistant** — Sep 2025
*DATA Lab at Northeastern University*

**Research Assistant** — Sep 2023 - Aug 2025
*Complex Analytics & Scalable Processing Research Lab (CASP) at Boston University*

## Research Projects

**Updatable and Scalable GPU Approximate Nearest Neighbor Index** — May 2025 (On-going)
*Advisor: Prashant Pandey, Northeastern University*

- Proposed and implemented a GPU vector index Jasper that is both updatable and can be scale outside GPU memory, enabling fast approximated nearest neighbor search with billion scale large data size.
- Achieving faster construction/query speed and smaller space usage compare to state-of-the-art library like cuvs.

**A Disaggregated Stream Processing System for Heterogeneous Environments** — Jan 2024 - Aug 2025
*Advisor: Prof. Vasiliki Kalavri, Boston University. Prof. Ioannis Liagouris, Boston University*

- Proposed a fully-disaggregated design for data stream processing system that decouples core data-plane and control-plane services, enabling non-blocking online reconfiguration without pausing the computation.
- Implemented a distributed streaming processor runtime that is able to support Flink-like API for dataflow definition. Developed a full test suite and a metric collection system to examine the performance of our runtime both in local and remote cloud environment. Integrated the metric package with Prometheus to allow developers to observe real-time telemetry.

**Automatic Task Placement for Apache Flink** — Oct 2023 - May 2024
*Advisor: Prof. Vasiliki Kalavri, Boston University*

- Implemented a task placement controller *CAPSys* for Apache Flink to efficiently delegate tasks to physical resources for data stream processing.
- Designed an adaptive task placement policy that ensures compute-intensive, I/O-intensive, and network-intensive tasks are balanced across available resources. When benchmarked against the state-of-the-art work, we achieved orders of magnitude lower computing time and up to 6× higher throughput with fixed resources.

## Teaching

Teaching Assistant | GRS CS 630 Graduate Algorithm | Boston University | Fall 2024
Course Grader | CAS CS 320 Concept Of Programming Language | Boston University | Fall 2022